

**Convention:** Throughout this discussion a feasible direction  $\mathbf{d}$  at a point is by definition taken to be a nonzero vector, although there is no significant harm even if assumed otherwise. Sometimes I may have forgotten to explicitly write it.

**Notation:**  $\nabla^2 f(\mathbf{x}) = H(\mathbf{x})$  is the Hessian matrix of  $f$  at  $\mathbf{x}$ .

$\nabla f(\mathbf{x})$  is the gradient **row** vector ( or the vector of partial derivatives written as a row vector).

## Nonlinear Programming

Let  $f$  be a real valued function defined on  $\Omega \subseteq \mathbb{R}^n$ .

The problem is to minimize or maximize  $f(\mathbf{x})$  subject to  $\mathbf{x} \in \Omega$ , where  $f$  need not be a linear function.

Throughout the discussion we will assume that  $\Omega \subseteq \mathbb{R}^n$ , for some  $n$ .

**Definition 1:** A point (or an element)  $\mathbf{x}^* \in \Omega$  is called a local minimum (maximum) of  $f$  if there exists an  $\epsilon > 0$ , such that

$\mathbf{x} \in \Omega$  and  $\|\mathbf{x} - \mathbf{x}^*\| < \epsilon$  implies  $f(\mathbf{x}^*) \leq f(\mathbf{x})$  ( $f(\mathbf{x}^*) \geq f(\mathbf{x})$ ).

**Definition 2:** A point  $\mathbf{x}^* \in \Omega$  is called a global minimum (maximum) of  $f$  if  $f(\mathbf{x}^*) \leq f(\mathbf{x})$  ( $f(\mathbf{x}^*) \geq f(\mathbf{x})$ ) for all  $\mathbf{x} \in \Omega$ .

**Definition 3:** A vector  $\mathbf{d} \in \mathbb{R}^n$ ,  $\mathbf{d} \neq \mathbf{0}$  is said to be a feasible direction at  $\mathbf{x}^* \in \Omega$ , if there exists a  $c > 0$  such that for all  $t$ ,  $0 \leq t \leq c$ ,  $\mathbf{x}^* + t\mathbf{d} \in \Omega$ .

**Example 1:** Let  $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$ .

At  $[0, 0]^T$  if  $\mathbf{d} = [d_1, d_2]^T$  is a feasible direction then  $d_1 \geq 0$  and  $d_2 \geq 0$ .

At  $[0, \frac{1}{2}]^T$  if  $\mathbf{d}$  is a feasible direction then  $d_1 \geq 0$  but  $d_2$  can be any real number.

At  $[\frac{1}{2}, 0]^T$  if  $\mathbf{d}$  is a feasible direction then  $d_2 \geq 0$  but  $d_1$  can be any real number.

At  $[\frac{1}{2}, \frac{1}{2}]^T$  any  $\mathbf{d} \in \mathbb{R}^2$  will be a feasible direction.

**Remark 1:** If  $\mathbf{x}^*$  is an interior point of  $\Omega$  then any  $\mathbf{d} \in \mathbb{R}^n$ ,  $\mathbf{d} \neq \mathbf{0}$  is a feasible direction at  $\mathbf{x}^*$ .

### *First order necessary conditions for a point to be a local minimum*

The results obtained in this section is based on **first order approximation** of the function  $f$  near the local minimum point  $\mathbf{x}^*$ .

Throughout this discussion we will assume  $\Omega \subseteq \mathbb{R}^n$  and  $\mathbf{x}^*, \mathbf{d}$  are elements of  $\mathbb{R}^n$  for some  $n \in \mathbb{N}$ .

**Theorem 1:** Let  $f : \Omega \rightarrow \mathbb{R}$  be a **continuously differentiable function** (that is, the first order partial derivatives of  $f$  exists and are continuous as functions from  $\Omega$  to  $\mathbb{R}$ ). If  $\mathbf{x}^*$  is a **local minimum point** then for any **feasible direction**  $\mathbf{d}$  at  $\mathbf{x}^*$ ,

$$\nabla f(\mathbf{x}^*)\mathbf{d} \geq 0,$$

where  $\nabla f(\mathbf{x}^*)$ , the gradient vector of  $f$  at  $\mathbf{x}^*$  is written as a row vector (the components of  $\nabla f(\mathbf{x}^*)$  are the first order partial derivatives of  $f$  at  $\mathbf{x}^*$ ) and  $\mathbf{d} \in \mathbb{R}^n$  is a column vector.

**Proof:** Let  $\mathbf{x}^*$  be a local minimum and let  $\mathbf{d}$  be a feasible direction at  $\mathbf{x}^*$ .

Let  $g(t) = f(\mathbf{x}(t))$ , where  $\mathbf{x}(t) = \mathbf{x}^* + t\mathbf{d}$ .

Since  $f$  is continuously differentiable throughout  $\Omega$  and  $\mathbf{d}$  is a **feasible direction** at  $\mathbf{x}^*$

$\lim_{h \rightarrow 0} \frac{f(\mathbf{x}^* + h\mathbf{d}) - f(\mathbf{x}^*)}{h}$  exists.

Since  $g(h) - g(0) = f(\mathbf{x}^* + h\mathbf{d}) - f(\mathbf{x}^*)$ ,  $\lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h}$  also exists and

$$g'(0) = \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} = \lim_{h \rightarrow 0} \frac{f(\mathbf{x}^* + h\mathbf{d}) - f(\mathbf{x}^*)}{h} = \nabla f(\mathbf{x}^*)\mathbf{d}.$$

Since  $g'(0)$  exists and we look at the first order Taylor's approximation of  $g$  around  $t = 0$ ,

$$g(t) = g(0) + tg'(0) + o(t), \text{ where } \lim_{t \rightarrow 0} \frac{o(t)}{t} = 0. \quad (**)$$

If we take  $0 < t \leq c$ , then  $(**)$  gives,

$$f(\mathbf{x}^* + t\mathbf{d}) = f(\mathbf{x}^*) + t\nabla f(\mathbf{x}^*)\mathbf{d} + o(t).$$

Since for  $t$  sufficiently small,

$$|t\nabla f(\mathbf{x}^*)\mathbf{d}| \geq |o(t)| \text{ (provided } \nabla f(\mathbf{x}^*)\mathbf{d} \neq 0),$$

if  $\nabla f(\mathbf{x}^*)\mathbf{d} < 0$ ,  $f(\mathbf{x}^* + t\mathbf{d}) < f(\mathbf{x}^*)$  for all  $t > 0$  sufficiently small, which contradicts that  $\mathbf{x}^*$  minimizes  $f$  locally.

**Theorem 2 :** Let  $f : \Omega \rightarrow \mathbb{R}$  be a **continuously differentiable** function. Let  $\mathbf{x}^*$  be an **interior point** of  $\Omega$ . If  $\mathbf{x}^*$  is a **local minimum point** of  $f$  then

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

**Proof:** Follows from **Theorem 1**, by taking  $\mathbf{d} = -(\nabla f(\mathbf{x}^*))^T$ . Since  $\mathbf{x}^*$  is an interior point we know that every  $\mathbf{d} \in \mathbb{R}^n, \mathbf{d} \neq \mathbf{0}$  is a feasible direction at  $\mathbf{x}^*$ .

### Second order necessary conditions for a point to be a local minimum

The following conditions are obtained by considering second order approximation of the function  $f$  near the local minimum point  $\mathbf{x}^*$ .

**Theorem 3:** Let  $f : \Omega \rightarrow \mathbb{R}$  be **twice continuously differentiable** (that is all the second order partial derivatives of  $f$  (given by  $\frac{\partial^2 f}{\partial x_j \partial x_i}$ ) exists and are continuous as functions from  $\Omega$  to  $\mathbb{R}$ ).

If  $\mathbf{x}^*$  is a **local minimum** of  $f$  then for any **feasible direction**  $\mathbf{d}$  at  $\mathbf{x}^*$

1.  $\nabla f(\mathbf{x}^*)\mathbf{d} \geq 0$ .
2. If  $\nabla f(\mathbf{x}^*)\mathbf{d} = 0$ , then  $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} \geq 0$ .

**Note:** The matrix  $\nabla^2 f$  (also denoted by  $H$ ) is called the **Hessian matrix** of  $f$ ,

$(\nabla^2 f(\mathbf{x}^*))_{ij} = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}^*) = \left[ \frac{\partial^2 f}{\partial x_j \partial x_i} \right]_{\mathbf{x}^*}$ . Since  $f$  **twice continuously differentiable** the **Hessian matrix**  $\nabla^2 f$  is a symmetric matrix for all  $\mathbf{x} \in \Omega$ .

**Proof:** Let  $\mathbf{d}$  be a **feasible direction** at  $\mathbf{x}^*$ . That  $\mathbf{x}^*$  satisfies condition 1 is already shown in Theorem 1.

As before, take  $g(t) = f(\mathbf{x}(t))$ , where  $\mathbf{x}(t) = \mathbf{x}^* + t\mathbf{d}$ .

Since  $f$  is **twice continuously differentiable**  $g''(0)$  exists. Also the **second order Taylor's approximation** of  $g$  around  $t = 0$  gives,

$$g(t) = g(0) + tg'(0) + \frac{t^2}{2!}g''(0) + o(t^2), \quad (**)$$

Since  $g'(t) = (\nabla f(\mathbf{x}^* + t\mathbf{d}))\mathbf{d} = \sum_i (\frac{\partial f}{\partial x_i})(\mathbf{x}^* + t\mathbf{d})d_i$ ,

$$g'(t) = \sum_i h_i(t)d_i,$$

$$\text{where } h_i(t) = \frac{\partial f}{\partial x_i}(\mathbf{x}^* + t\mathbf{d}).$$

$$\text{Hence } g''(t) = \sum_i h'_i(t)d_i,$$

$$\text{where } h'_i(t) = ((\nabla \frac{\partial f}{\partial x_i})(\mathbf{x}^* + t\mathbf{d}))\mathbf{d} = \sum_j \frac{\partial}{\partial x_j}(\frac{\partial f}{\partial x_i})(\mathbf{x}^* + t\mathbf{d})d_j = \sum_j (\frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}^* + t\mathbf{d}))d_j.$$

$$\text{Hence } g''(t) = \sum_i h'_i(t)d_i = \sum_i (\sum_j (\frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}^* + t\mathbf{d}))d_j)d_i,$$

$$\text{Hence } g''(0) = \sum_i (\sum_j (\frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}^*))d_j)d_i = \mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d},$$

where  $\nabla^2 f(\mathbf{x}^*) (= H(\mathbf{x}^*))$  is an  $n \times n$  matrix whose  $(i, j)$  th entry is given by  $\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}^*)$ .

Note that because we have assumed  $f$  to be **twice continuously differentiable** the matrix  $\nabla^2 f(\mathbf{x}^*)$  is a symmetric matrix.

Again from  $(**)$  we get,

$$f(\mathbf{x}^* + t\mathbf{d}) = f(\mathbf{x}^*) + t\nabla f(\mathbf{x}^*)\mathbf{d} + \frac{t^2}{2}\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} + o(t^2).$$

Since for sufficiently small  $t$ ,

$$|\frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d}| \geq |o(t^2)|, \text{ ( provided } \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \neq 0 \text{ )}$$

hence if  $\nabla f(\mathbf{x}^*) \mathbf{d} = 0$  and  $\mathbf{x}^*$  is a **local minimum** then it should satisfy the condition  $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$  if  $\nabla f(\mathbf{x}^*) \mathbf{d} = 0$ .

**Theorem 4:** Let  $f : \Omega \rightarrow \mathbb{R}$  be a twice continuously differentiable function and let  $\mathbf{x}^*$  be an interior point of  $\Omega$ . If  $\mathbf{x}^*$  is a local minimum of  $f$  then

1.  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ .
2.  $\nabla^2 f(\mathbf{x}^*)$  is positive semidefinite( defined later ).

**Proof:** Follows from the previous theorem and the fact that for an **interior point** of  $\Omega$ , every nonzero vector  $\mathbf{d} \in \mathbb{R}^n$  is a feasible direction.

Hence from **Theorem 3**,  $\nabla f(\mathbf{x}^*)(-\nabla f(\mathbf{x}^*))^T \geq 0$ , which implies  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ .

Applying the second order conclusion of **Theorem 3**, we get  $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$  for all  $\mathbf{d} \in \mathbb{R}^n$ .

**Definition:** A real symmetric matrix  $A$  is said to be positive semidefinite (negative semidefinite) if  $\mathbf{x}^T A \mathbf{x} \geq 0$  ( $\mathbf{x}^T A \mathbf{x} \leq 0$ ) for all  $\mathbf{x} \in \mathbb{R}^n$ .

**Definition:** A real symmetric matrix  $A$  is said to be positive definite (negative definite) if  $\mathbf{x}^T A \mathbf{x} > 0$  ( $\mathbf{x}^T A \mathbf{x} < 0$ ) for all **nonzero** vectors  $\mathbf{x} \in \mathbb{R}^n$ .

**Remark:** Note that in general a matrix satisfying the condition  $\mathbf{x}^T A \mathbf{x} \geq 0$  for all  $\mathbf{x} \in \mathbb{R}^n$  need not be symmetric for example  $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ .

**Theorem:** If  $A$  is a symmetric,  $n \times n$ , real matrix then the following statements are equivalent:

1.  $A$  is positive semidefinite.
2. All eigenvalues of  $A$  are nonnegative.
3. All principal minors of  $A$  are nonnegative.

**Definition:**  $\lambda$  is called an eigenvalue of an  $n \times n$  matrix  $A$  if there exists an  $\mathbf{x} \in \mathbb{C}^n, \mathbf{x} \neq \mathbf{0}$  ( that is atleast one component of  $\mathbf{x}$  is nonzero) such that  $A\mathbf{x} = \lambda\mathbf{x}$ .

For example the  $\mathbf{0}$  matrix has all  $n$  eigenvalues equal to 0, the identity matrix  $I_n$  has all  $n$  eigenvalues equal to 1 and for an upper triangular matrix the diagonal entries are its eigenvalues.

**Definition:** If  $A$  is an  $n \times n$  matrix and  $\alpha \subseteq \{1, \dots, n\}, \beta \subseteq \{1, \dots, n\}$  then  $A[\alpha, \beta]$  is the (sub)matrix obtained from  $A$  by deleting all the rows of  $A$  which do not belong to  $\alpha$  and by deleting all the columns of  $A$  which do not belong to  $\beta$ .

If  $\alpha = \beta$  then  $A[\alpha, \alpha]$  is called a principal submatrix of  $A$  and  $\det A[\alpha, \alpha]$  is called a principal minor of  $A$ .

For example if  $\alpha = \beta = \{i\}$  where  $i \in \{1, \dots, n\}$  then  $A[\alpha, \alpha] = [a_{ii}]$  and  $\det A[\alpha, \alpha] = a_{ii}$ , the  $i$  th diagonal entry.

If  $\alpha = \beta = \{i, j\}$  where  $i, j \in \{1, \dots, n\}$  and  $i < j$  then  $A[\alpha, \alpha] = \begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$ .

If  $\alpha = \{1, \dots, n\}$  then  $A[\alpha, \alpha] = A$  and  $\det A[\alpha, \alpha] = \det(A)$ .

**Remark:** A nonsingular (nonzero determinant) positive semidefinite matrix is positive definite.

In the following examples there is a slight abuse of notation. Instead of writing  $f([x_1, x_2]^T)$ , to avoid cumbersome notation, I have written it as  $f(x_1, x_2)$ .

**Example 1 revisited:** Consider the following problem:

Minimize  $f(x_1, x_2) = x_1^3 - x_1^2 x_2 + 2x_2^2$

subject to  $x_1 \geq 0, x_2 \geq 0$ , hence  $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$ .

To search for points which might be local minimum points of  $f$  we initially search for points in  $\Omega$  which satisfy first order necessary conditions for a local minimum. The points which obviously satisfy first order necessary conditions for a local minimum are the points at which the gradient vector vanishes.

Hence we try to find  $\mathbf{x}$  at which  $\nabla f(\mathbf{x}) = 0$ .

$\nabla f(\mathbf{x}) = (3x_1^2 - 2x_1 x_2, -x_1^2 + 4x_2) = [0, 0]$  has two solutions  $x_1 = 0, x_2 = 0$  and  $x_1 = 6, x_2 = 9$ .

Here  $[6, 9]^T$  an interior point of the feasible region  $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$  satisfies the first order necessary conditions for a local minimum.

But since  $\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 6x_1 - 2x_2 & -2x_1 \\ -2x_1 & 4 \end{pmatrix}$ .

At  $\mathbf{x}^* = [6, 9]^T$ ,  $\nabla^2 f(\mathbf{x}^*) = \begin{pmatrix} 18 & -12 \\ -12 & 4 \end{pmatrix}$  is not positive semidefinite.

Hence  $\mathbf{x}^* = [6, 9]^T$  is not a local minimum point of  $f$ .

Hence the first order necessary conditions are **necessary** but **not sufficient** for a point to be a local minimum.

At  $\mathbf{x}^* = [0, 0]^T$  a  $\mathbf{d} \neq \mathbf{0}$  is a feasible direction if and only if  $d_1 \geq 0$  and  $d_2 \geq 0$ .

Since  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ ,  $\nabla f(\mathbf{x}^*)\mathbf{d} = 0$  for all  $\mathbf{d}$ .

Since  $\nabla^2 f(\mathbf{x}^*) = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$ ,  $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} = 4d_2^2 \geq 0$  for all  $\mathbf{d}$ .

Since  $[0, 0]^T$  satisfies both the first and second order necessary conditions, we can only say that  $\mathbf{x}^* = [0, 0]^T$  can be a candidate for local minimum, but since these are only necessary conditions we cannot conclude from previous calculations that  $[0, 0]^T$  is indeed a local minimum of  $f$  in  $\Omega$ .

**Exercise:** Check that  $[0, 0]^T$  is a local minimum point of  $f$  in **Example 1**.

**Solution:** Note that  $f(x_1, x_2) = x_1^3 - x_1^2 x_2 + 2x_2^2 = x_1^2(x_1 - x_2) + 2x_2^2$  can take values  $< 0$  only when either  $x_2 = 0$  and  $x_1 < 0$  or when  $x_2 \neq 0$  and  $x_2 > x_1$ . But  $[-1, 0]^T$  is not a feasible direction at  $\mathbf{x}^* = [0, 0]^T$  for  $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$ .

For  $x_2 \neq 0$  and  $x_2 > x_1$  sufficiently small, that is  $|x_1|, |x_2| < 1$ , clearly  $x_1^2(x_1 - x_2) + 2x_2^2 \geq 0$ . Hence  $[0, 0]^T$  is a local minimum of  $f$ .

It is quite clear that for any  $f$ , of the form  $f(x_1, x_2) = x_1^3 - x_1^2 x_2 + cx_2^2, c > 0$ ,  $\mathbf{x}^* = [0, 0]^T$  will be a local minimum point of  $f$  for the domain given in **Example 1**.

However if we take  $c = 0$  in the above expression, then  $f(x_1, x_2) = x_1^3 - x_1^2 x_2$ , then one can check that  $\nabla^2 f(\mathbf{x}^*) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ , is a positive semidefinite matrix, but  $\mathbf{x}^* = [0, 0]^T$  is not a local optimum point.

Hence the **first order** and the **second order** necessary conditions are **necessary** but **not sufficient** for a point  $\mathbf{x}^*$  to be a **local minimum**.

**Exercise:** Will  $[0, 0]^T$  be a local minimum point of  $f$  given in **Example 1** if  $\Omega = \mathbb{R}^2$  ?

**Example 2 :** Consider the following problem:

Minimize  $f(x_1, x_2) = x_1^2 - x_1 + x_2 + x_1 x_2$

subject to  $x_1 \geq 0, x_2 \geq 0$ .

$f$  is a twice continuously differentiable function.

At  $[\frac{1}{2}, 0]^T$ ,  $\frac{\partial f}{\partial x_1} = 2x_1 - 1 + x_2 = 0$

$$\frac{\partial f}{\partial x_2} = 1 + x_1 = \frac{3}{2}.$$

If  $\mathbf{d}$  is a feasible direction at  $[\frac{1}{2}, 0]^T$ , then  $d_2$  has to be nonnegative.

Hence  $\nabla f(\mathbf{x})|_{[\frac{1}{2}, 0]^T} \mathbf{d} = \frac{3}{2}d_2 \geq 0$  for any feasible direction  $\mathbf{d}$ .

Hence the first order necessary conditions for  $\mathbf{x}^* = [\frac{1}{2}, 0]^T$  to be a locally minimum point is satisfied.

Also if  $\nabla f(\mathbf{x}^*)\mathbf{d} = 0$ , then  $d_2 = 0$ , and for all such  $\mathbf{d}$

$$\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} = 2d_1^2 \geq 0.$$

Hence the second order necessary conditions for  $\mathbf{x}^*$  to be locally minimum is also satisfied.

One can easily check that  $f$  has a global minimum at  $x_1 = \frac{1}{2}, x_2 = 0$ .

### Sufficient conditions for a local minima :

**Theorem 4 :** Let  $f : \Omega \rightarrow \mathbb{R}$  be a **twice continuously differentiable** function. Let  $\mathbf{x}^*$  be an interior point of  $\Omega$ . If  $\mathbf{x}^*$  satisfies the following conditions

$$1. \nabla f(\mathbf{x}^*) = \mathbf{0}$$

$$2. \nabla^2 f(\mathbf{x}^*) \text{ is positive definite,}$$

then  $\mathbf{x}^*$  is a **local minimum point** of  $f$ .

**Proof:** Since  $\mathbf{x}^*$  is an interior point of  $\Omega$ , if  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ , then by Taylor's second order approximation formula for  $f$  near  $\mathbf{x}^*$ , we get

$$f(\mathbf{x}^* + t\mathbf{d}) = f(\mathbf{x}^*) + \frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(t^2), \text{ for all } \mathbf{d} \in \mathbb{R}^n \text{ and all } t > 0 \text{ sufficiently small.}$$

For  $t$  small,  $|\frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d}| > |o(t^2)|$  (if  $\mathbf{d} \neq \mathbf{0}$ ).

Hence  $f(\mathbf{x}^* + t\mathbf{d}) > f(\mathbf{x}^*)$  for all  $t$  sufficiently small. The above proof is not enough to conclude that  $\mathbf{x}^*$  is a **local minimum point** of  $f$ .

The actual proof is given below but you may skip it if you find it difficult.

### Optional reading:

$$f(\mathbf{x}^* + \mathbf{d}) = f(\mathbf{x}^*) + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(\|\mathbf{d}\|^2).$$

Since  $\nabla^2 f(\mathbf{x}^*)$  is real, symmetric ( By spectral theorem for real symmetric matrices), there exists a real matrix  $U$ , orthogonal ( that is,  $U^T U = U U^T = I$ ) such that

$$\nabla^2 f(\mathbf{x}^*) = U^T \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{pmatrix} U, \text{ where } \lambda_i \text{'s are the eigenvalues of } \nabla^2 f(\mathbf{x}^*).$$

$$\Rightarrow \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} = \mathbf{y}^T \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{pmatrix} \mathbf{y}, \text{ where } \mathbf{y} = U\mathbf{d}.$$

$$\Rightarrow \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} = \sum_i \lambda_i y_i^2, \text{ where } y_i \text{ s the } i \text{ th component of } \mathbf{y}. \text{ (multiply and check).}$$

$$\Rightarrow \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq \lambda_{\min} \sum_i y_i^2 = \lambda_{\min} \mathbf{y}^T \mathbf{y} = \lambda_{\min} \mathbf{d}^T \mathbf{d}$$

(  $\mathbf{y}^T \mathbf{y} = \mathbf{d}^T U^T U \mathbf{d} = \mathbf{d}^T \mathbf{d} = \|\mathbf{d}\|^2$  since  $U^T U = I$ ).

Hence  $f(\mathbf{x}^* + \mathbf{d}) \geq f(\mathbf{x}^*) + \lambda_{\min} \|\mathbf{d}\|^2 + o(\|\mathbf{d}\|^2)$ .

For  $\|\mathbf{d}\| > 0$  sufficiently small we have

$|\lambda_{\min} \|\mathbf{d}\|^2| > |o(\|\mathbf{d}\|^2)|$  hence  $f(\mathbf{x}^* + \mathbf{d}) > f(\mathbf{x}^*)$  for all  $\|\mathbf{d}\| > 0$ , sufficiently small.

**Remark:** Since **maximizing**  $f$  is same as **minimizing**  $-f$ , all the previous theorems have corresponding analogues for a **maximization problem** with some obvious changes. For example  $\leq$  conditions in the results are replaced by  $\geq$  conditions and with positive semidefinite (or positive

definite ) matrices in the results are appropriately replaced by negative semidefinite matrices (or negative definite matrices).

**Definition 4:** A real valued function  $f$  defined on a **convex set**  $\Omega \subseteq \mathbb{R}^n$  is said to be a **convex function** on  $\Omega$  if for all  $\mathbf{x}, \mathbf{y} \in \Omega$  and all  $0 \leq \alpha \leq 1$ ,  
 $f(\alpha \mathbf{x} + (1 - \alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y})$ .

**Definition 5:** An  $f : \Omega \rightarrow \mathbb{R}$  is said to be a concave function if  $-f$  is a convex function.

**Theorem 1 :** If  $f$  is a **convex function** on  $\Omega$  (a convex set), then the set  $S = \{\mathbf{x} \in \Omega : f(\mathbf{x}) \leq c\}$  is a convex set (for all real  $c$ ).

**Proof:** Exercise.

**Theorem 2:** Let  $f$  be a **continuously differentiable** function defined on a convex set,  $\Omega \subseteq \mathbb{R}^n$ , then  $f$  is **convex** on  $\Omega$  if and only if  $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$  for all  $\mathbf{x}, \mathbf{y} \in \Omega$ .

**Proof:** Let  $f : \Omega \rightarrow \mathbb{R}$  be a convex function. Then for all  $\mathbf{x}, \mathbf{y} \in \Omega$  and all  $0 \leq \alpha \leq 1$ ,  
 $f(\alpha \mathbf{y} + (1 - \alpha)\mathbf{x}) \leq \alpha f(\mathbf{y}) + (1 - \alpha)f(\mathbf{x})$ .

For all  $\alpha > 0$ , sufficiently small,  

$$\frac{f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\alpha} \leq f(\mathbf{y}) - f(\mathbf{x})$$
  
 Letting  $\alpha \rightarrow 0$  we get  
 $\nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x})$ .

To show the converse,  
 let  $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$  for all  $\mathbf{x}, \mathbf{y} \in \Omega$ . (\*\*)  
 Fix  $\mathbf{x}, \mathbf{y} \in \Omega$ , and let  $\mathbf{z}$  be a point in between and on the straight line segment joining  $\mathbf{x}$  and  $\mathbf{y}$ .  
 That is  $\mathbf{z} = \alpha \mathbf{x} + (1 - \alpha)\mathbf{y}$  for some  $0 \leq \alpha \leq 1$ . From (\*\*) we get  
 $f(\mathbf{x}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})(\mathbf{x} - \mathbf{z})$  and  
 $f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})(\mathbf{y} - \mathbf{z})$ .  
 By multiplying the first equation by  $\alpha$ , the second by  $(1 - \alpha)$  and adding the two equations we get,  
 $\alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})(\alpha \mathbf{x} + (1 - \alpha)\mathbf{y} - \mathbf{z})$ .  
 Since  $\mathbf{z} = \alpha \mathbf{x} + (1 - \alpha)\mathbf{y}$  we get  
 $\alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}) \geq f(\alpha \mathbf{x} + (1 - \alpha)\mathbf{y})$ .  
 But the above condition is difficult to verify. The following result gives an easy way to the convexity of a function provided that the function satisfies certain conditions.

**Theorem 3 :** Let  $f$  be a **twice continuously differentiable** function on a convex set  $\Omega$  (let  $\Omega$  be such that it has atleast one interior point), then  $f$  is convex on  $\Omega$  if and only if for all  $\mathbf{x} \in \Omega$ ,  $\nabla^2 f(\mathbf{x})$  is **positive semidefinite**.

**Proof:** For those interested in the proof, refer to Luenberger( Page 196, Third Edition) or Bazaara Sherali Shetty (page 92, Second edition).

**Revisiting Example 1 :** Let  $f(x_1, x_2) = x_1^3 - x_1^2 x_2 + 2x_2^2$  be defined on  $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$ .

Since  $\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 6x_1 - 2x_2 & -2x_1 \\ -2x_1 & 4 \end{pmatrix}$ , at  $x_1 = 1, x_2 = 3$ ,  $\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 0 & -2 \\ -2 & 4 \end{pmatrix}$   
 is clearly not positive semidefinite. Hence  $f$  is not a convex function on  $\Omega$ .

**Theorem 4:** Let  $f : \Omega \rightarrow \mathbb{R}$  be a **continuously differentiable** function. If  $f$  is **convex** on  $\Omega$ , then  $\mathbf{x}^*$  is a **global minimum** of  $f$  if and only if for all **feasible directions**  $\mathbf{d}$  at  $\mathbf{x}^*$   $\nabla f(\mathbf{x}^*)\mathbf{d} \geq 0$ .

**Proof:** Since the **only if** part is already shown before, we have to only show the **if** part.

Let  $\mathbf{x}^* \in \Omega$  satisfy  $\nabla f(\mathbf{x}^*)\mathbf{d} \geq 0$  for all feasible  $\mathbf{d}$  at  $\mathbf{x}^*$ .

Since  $f$  is convex  $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$  for all  $\mathbf{x}, \mathbf{y} \in \Omega$ . (1)

Hence for all  $\mathbf{y} \in \Omega$ ,  $f(\mathbf{y}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*)$ .

Since  $\Omega$  is convex,  $\mathbf{x} = \alpha\mathbf{y} + (1 - \alpha)\mathbf{x}^* = \mathbf{x}^* + \alpha(\mathbf{y} - \mathbf{x}^*)$  belongs to  $\Omega$ , for all  $0 \leq \alpha \leq 1$ , hence  $(\mathbf{y} - \mathbf{x}^*)$  is a feasible direction at  $\mathbf{x}^*$  and  $\nabla f(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*) \geq 0$ .

Hence  $f(\mathbf{y}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*) \geq f(\mathbf{x}^*)$ .

Since  $\mathbf{y} \in \Omega$  was arbitrary,  $\mathbf{x}^*$  is a global minimum of  $f$ .

**Corollary 4:** Let  $f : \Omega \rightarrow \mathbb{R}$  be a **continuously differentiable** function. If  $f$  is **convex** on  $\Omega$  and  $\mathbf{x}^*$  an **interior point** of  $\Omega$ , then  $\mathbf{x}^*$  is a **global minimum** for  $f$  if and only if

$\nabla f(\mathbf{x}^*) = \mathbf{0}$ .

**Proof:** Follows from the previous result.

That the above result is not necessarily true if  $f$  is not convex as you have already seen in **Example 1**.

**Remark :** Since **minimizing**  $f$  is same as **maximizing**  $-f$ , all the previous theorems for **minimizing a convex function** have corresponding analogues for **maximizing a concave function**.

**Theorem 5:** Let  $f : \Omega \rightarrow \mathbb{R}$  be a **convex function**, then the following statements are true.

1. Let  $S$  be the collection of all  $\mathbf{x}$ 's where  $f$  attains its **minimum value** (that is, the set of all optimal solutions of  $f$  for a minimization problem). Then  $S$  is a **convex set**, or in other words the set  $S = \{\mathbf{x} : f(\mathbf{x}) \leq f(\mathbf{y}) \text{ for all } \mathbf{y} \in \Omega\}$ , is a **convex set**.
2. If  $\mathbf{x}^*$  is a **local minimum** point of  $f$  then it is also a **global minimum** point of  $f$ .

**Proof:**

1. If  $f$  does not have a minimum then the above result is vacuously true.

If  $f$  takes a minimum value then let  $a = \min_{\mathbf{x} \in \Omega} \{f(\mathbf{x})\}$ .

Since  $f$  is a convex function,  $S_1 = \{\mathbf{x} : f(\mathbf{x}) \leq a\}$  is convex, by Theorem 1.

Note that  $S_1 = S$ .

2. To show that a local minimum of  $f$  is a global minimum of  $f$ .

If not, then let  $\mathbf{x}^*$  be a local minimum point of  $f$  and let there exist a  $\mathbf{y} \in \Omega$  such that  $f(\mathbf{y}) < f(\mathbf{x}^*)$ .

Join  $\mathbf{x}^*$  and  $\mathbf{y}$  by a straight line.

Since  $\Omega$  is a convex set, the straight line segment joining  $\mathbf{x}^*$  and  $\mathbf{y}$  lies entirely in  $\Omega$ .

Since  $f$  is a convex function, for all  $0 < \alpha \leq 1$ ,

$f((1 - \alpha)\mathbf{x}^* + \alpha\mathbf{y}) \leq (1 - \alpha)f(\mathbf{x}^*) + \alpha f(\mathbf{y}) < f(\mathbf{x}^*)$ .

This contradicts that  $\mathbf{x}^*$  is a local minimum point of  $f$ .

**Remark:** A natural question would be whether the conclusions of Theorem 5 holds good when **maximizing a convex function**. The answer however is a **NO**.

Take  $f(x) = x^2$ ,  $-1 \leq x \leq 2$ .

From previous discussions however it is clear that the above result is true if you are **maximizing**  $-f$  or **maximizing** a **concave** function.

**Remark:** We had seen while **minimizing** or **maximizing** a **linear** function over a **polyhedral** set, the extremum was attained in at least one **extreme point**. An extreme point of a polyhedral set, ( more generally convex set) is one which cannot be written as a strict convex combination of two distinct points of that set).

But in the problem of **minimizing** a **convex** function over a **polyhedral** set, the only minimum may be attained at an **interior point** of  $\Omega$ .

For example take  $f(x) = x^2$ ,  $-1 \leq x \leq 1$ , the only minimum of this function on the interval  $[-1, 1]$  which is a convex set is attained at  $x = 0$  which is an interior point of  $[-1, 1]$ .

However when maximizing a convex function over a closed and bounded convex set the maximum is attained at an extreme point as the following theorem will illustrate.

**Theorem 6:** Let  $f$  be a **convex** function defined on a **closed** and **bounded convex** set  $\Omega$  (so it has atleast one extreme point), then there exists an **extreme point** of  $\Omega$ , where  $f$  takes its **maximum** value. (In case you are interested to know the proof refer to Luenberger page 198, Third Edition).